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Segmented variable fan outlet guide vane with cam assembly and pass through actuation mechanisms

Abstract

A fan assembly for a gas turbine engine includes a fan duct, a fan, and an outlet guide vane assembly located in the fan duct axially downstream of the fan. The outlet guide vane assembly includes a first variable leading edge outlet guide vane that extends radially relative to the central axis and includes a leading edge portion and a fixed aft portion, the leading edge portion including a tip segment configured to rotate about a leading edge pitch axis and a hub segment located radially inward of and separate from the tip segment, the hub segment configured to independently rotate about the leading edge pitch axis relative to the tip segment. The assembly further includes a single actuation assembly includes a cam rod and cams arranged thereon that is configured to rotate the tip and hub segments via the cams.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Application Ser. No. 63/452,883, filed on Mar. 17, 2023, the entire disclosure of which is incorporated herein by reference.

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to gas turbine engines, and more specifically to fan assemblies of gas turbine engines.

BACKGROUND

(2) Gas turbine engines are used to power aircraft, watercraft, power generators, and the like. Gas turbine engines typically include an engine core having a compressor, a combustor, and a turbine. The compressor compresses air drawn into the engine and delivers high pressure air to the combustor. In the combustor, fuel is mixed with the high pressure air and is ignited. Products of the combustion reaction in the combustor are directed into the turbine where work is extracted to drive the compressor and, sometimes, an output shaft. Left-over products of the combustion are exhausted out of the turbine and may provide thrust in some applications.

(3) Gas turbine engines also typically include a fan positioned within an inlet duct of the gas turbine engine. The fan includes rotating blades that force air into the compressor section of the engine, as well as potentially providing additional thrust via forcing air around the engine core through bypass ducts. The fan blades may experience various operability issues due to factors such as variations in the intake airflow and pressure fluctuations within the inlet and the bypass ducts.

SUMMARY

(4) The present disclosure may comprise one or more of the following features and combinations thereof.

(5) A fan assembly for a gas turbine engine according to the present disclosure includes a fan duct arranged circumferentially around a central axis, a fan comprising a plurality of fan blades that extend radially outward relative to the central axis and that are adapted to rotate about the central axis to force fan exit air toward an aft end of the fan duct, and an outlet guide vane assembly located in the fan duct axially downstream of the fan and configured to adjust a direction of incoming fan exit air received from the plurality of fan blades.

(6) The outlet guide vane assembly includes a first variable leading edge outlet guide vane that extends radially relative to the central axis and includes a leading edge portion and a fixed aft portion, the leading edge portion including a first tip segment configured to rotate about a leading edge pitch axis and a first hub segment located radially inward of and separate from the first tip segment, the first hub segment configured to independently rotate about the leading edge pitch axis relative to the first tip segment, a first actuation assembly including a first cam rod, a first cam selectively coupled to the first cam rod, and a second cam selectively coupled to the first cam rod and radially spaced apart from the first cam, wherein the first cam rod is configured to be selectively rotated so as to rotate the first cam and the second cam, wherein rotation of the first cam causes rotation of the first tip segment about the leading edge pitch axis, wherein rotation of the second cam causes rotation of the first hub segment about the leading edge pitch axis, and wherein the first and second cams are configured to be selectively clocked to unique rotational positions on the first cam rod such that rotation of the first and second cams causes the first tip segment to be rotated to a first pitch angle relative to the incoming fan exit air and the first hub segment to be rotated to a second angle relative to the incoming fan exit air.

(7) In some embodiments, the first actuation assembly is configured to rotate the first tip segment and the first hub segment to the first pitch angle and the second pitch angle which is different than the first pitch angle.

- (8) In some embodiments, the first actuation assembly further includes a first cam actuation rod rotatably coupled to the first tip segment and extending axially aft therefrom toward the first cam, and a second cam actuation rod rotatably coupled to the first hub segment extending axially aft therefrom toward the second cam.
- (9) In some embodiments, an axially aft end of the first cam actuation rod engages an axially forward facing surface of the first cam such that rotation of the first cam in a first rotational direction causes the axially forward facing surface to move the first cam actuation rod axially forward, and an axially aft end of the second cam actuation rod engages an axially forward facing surface of the second cam such that rotation of the second cam in the first rotational direction causes the axially forward facing surface to move the second cam actuation rod axially forward.
- (10) In some embodiments, the first cam actuation rod is rotatably coupled to the first tip segment at a point offset from the leading edge pitch axis in a circumferential direction such that axially forward movement of the first cam actuation rod rotates the first tip segment in the first rotational direction, and the second cam actuation rod is rotatably coupled to the first hub segment at a point offset from the leading edge pitch axis in the circumferential direction such that axially forward movement of the second cam actuation rod rotates the first hub segment in the first rotational direction.
- (11) In some embodiments, the first tip segment includes a first cam actuation rod receiving recess formed at least partially in an aft side of the first tip segment, the first cam actuation rod is rotatably mounted within the first cam actuation rod receiving recess, the first hub segment includes a second cam actuation rod receiving recess formed at least partially in an aft side of the first hub segment, and the second cam actuation rod is rotatably mounted within the second cam actuation rod receiving recess.
- (12) In some embodiments, the first actuation assembly is arranged radially outward of the fixed aft portion and includes a first actuation head, and the first cam rod is fixedly coupled to the first actuation head such that rotation of the first actuation head causes rotation of the first cam rod.
- (13) In some embodiments, the first actuation assembly further includes a first actuation arm coupled to the first actuation head and extending generally axially aft therefrom, pivoting of the first actuation arm about a rotation axis of the first actuation head causes rotation of the first actuation head which causes rotation of the first cam rod.
- (14) In some embodiments, the fixed aft portion includes a cam rod receiving cavity formed therethrough, and the first actuation head is circumferentially aligned with the fixed aft portion such that the first cam rod extends through the cam rod receiving cavity.
- (15) In some embodiments, the first actuation head is circumferentially offset from the fixed aft portion such that the first cam rod radially extends circumferentially adjacent to the fixed aft portion.
- (16) In some embodiments, the outlet guide vane assembly further includes a first annular ring extending circumferentially about the central axis, wherein an axially aft end of the first actuation arm is connected to the first annular ring, and circumferential movement of the first annular ring causes movement of the first actuation arm generally circumferentially relative to the first actuation head which causes the first actuation arm to pivot about the rotation axis of the first actuation head which causes rotation of the first actuation head which causes rotation of the first cam rod.
- (17) In some embodiments, the first variable leading edge outlet guide vane further includes a central segment arranged between the first tip segment and the first hub segment such that the first tip segment and the first hub segment are radially spaced apart.
- (18) In some embodiments, the central segment is coupled to and extends axially away from an axially forward side of the fixed aft portion.
- (19) According to a further aspect of the present disclosure, a fan assembly for a gas turbine engine includes a fan duct arranged circumferentially around a central axis, a fan adapted to rotate about the central axis to force fan exit air toward an aft end of the fan duct, and an outlet guide vane

assembly located in the fan duct axially downstream of the fan and configured to adjust a direction of incoming fan exit air received from the plurality of fan blades.

(20) The outlet guide vane assembly includes a first variable leading edge outlet guide vane including a leading edge portion and a fixed aft portion, the leading edge portion including a first tip segment configured to rotate about a leading edge pitch axis, and a first actuation assembly including a first cam rod and a first cam selectively coupled to the first cam rod, wherein the first cam rod is configured to be selectively rotated so as to rotate the first cam, wherein rotation of the first cam causes rotation of the first tip segment about the leading edge pitch axis, and wherein the first cam is configured to be selectively clocked to unique rotational positions on the first cam rod such that rotation of the first cam causes the first tip segment to be rotated to a first pitch angle relative to the incoming fan exit air.

(21) In some embodiments, the leading edge portion further includes a first hub segment located radially inward of and separate from the first tip segment, the first hub segment configured to independently rotate about the leading edge pitch axis relative to the first tip segment.

(22) In some embodiments, the first actuation assembly further includes a second cam selectively coupled to the first cam rod and radially spaced apart from the first cam, the first cam rod is configured to be selectively rotated so as to rotate the first cam and the second cam, and the second cam is configured to be selectively clocked to unique rotational positions on the first cam rod such that rotation of the first and second cams causes the first tip segment to be rotated to a first pitch angle relative to the incoming fan exit air and the first hub segment to be rotated to a second angle relative to the incoming fan exit air.

(23) In some embodiments, the first actuation assembly further includes a first cam actuation rod rotatably coupled to the first tip segment and extending axially aft therefrom toward the first cam, and a second cam actuation rod rotatably coupled to the first hub segment extending axially aft therefrom toward the second cam.

(24) In some embodiments, an axially aft end of the first cam actuation rod engages an axially forward facing surface of the first cam such that rotation of the first cam in a first rotational direction causes the axially forward facing surface to move the first cam actuation rod axially forward, and an axially aft end of the second cam actuation rod engages an axially forward facing surface of the second cam such that rotation of the second cam in the first rotational direction causes the axially forward facing surface to move the second cam actuation rod axially forward.

(25) In some embodiments, the first cam actuation rod is rotatably coupled to the first tip segment at a point offset from the leading edge pitch axis in a circumferential direction such that axially forward movement of the first cam actuation rod rotates the first tip segment in the first rotational direction, and the second cam actuation rod is rotatably coupled to the first hub segment at a point offset from the leading edge pitch axis in the circumferential direction such that axially forward movement of the second cam actuation rod rotates the first hub segment in the first rotational direction.

(26) According to a further aspect of the present disclosure, a method comprises arranging a fan duct circumferentially around a central axis, providing a fan comprising a plurality of fan blades that extend radially outward relative to the central axis and that are adapted to rotate about the central axis to force fan exit air toward an aft end of the fan duct, arranging an outlet guide vane assembly in the fan duct axially downstream of the fan and configured to adjust a direction of incoming fan exit air received from the plurality of fan blades, the outlet guide vane assembly including a first variable leading edge guide vane that extends radially relative to the central axis and includes a leading edge portion and a fixed aft portion, the leading edge portion including a first tip segment configured to rotate about a leading edge pitch axis and a first hub segment located radially inward of and separate from the first tip segment, the first hub segment configured to independently rotate about the leading edge pitch axis relative to the first tip segment, arranging a first actuation assembly relative to the first variable leading edge guide vane, the first actuation

assembly including a first cam rod, selectively coupling a first cam to the first cam rod, and selectively coupling a second cam to the first cam rod radially spaced apart from the first cam.

(27) The first cam rod is configured to be selectively rotated so as to rotate the first cam and the second cam, wherein rotation of the first cam causes rotation of the first tip segment about the leading edge pitch axis, wherein rotation of the second cam causes rotation of the first hub segment about the leading edge pitch axis, and wherein the first and second cams are configured to be selectively clocked to unique rotational positions on the first cam rod such that rotation of the first and second cams causes the first tip segment to be rotated to a first pitch angle relative to the incoming fan exit air and the first hub segment to be rotated to a second angle relative to the incoming fan exit air.

(28) These and other features of the present disclosure will become more apparent from the following description of the illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a cutaway view of a gas turbine engine that includes a fan assembly having a fan having plurality of fan blades that extend radially outward relative to the central axis, an engine core having a compressor, a combustor, and a turbine, and an outlet guide vane assembly located in a fan duct axially downstream of the plurality of fan blades that is configured to reduce the incidence between the outlet guide vane assembly and the fan exit air received from the fan blades and return the flow to generally axial flow;
- (2) FIG. 2 is a side cross-sectional view of the gas turbine engine of FIG. 1, showing the fan assembly including the plurality of fan blades, showing that the engine further includes an outer casing and an inner wall that define a fan duct passage through which the fan exit air flows, showing that the outlet guide vane assembly includes a first variable leading edge outlet guide vane having a first tip segment and a first hub segment configured to more closely match the vane to distorted fan exit air, and showing that the first tip and hub segments are hinged to each other;
- (3) FIG. 3 is a side cross-sectional view of the first variable leading edge outlet guide vane of FIG. 2, showing that the first variable leading edge outlet guide vane includes the first tip segment and the first hub segment, and showing that the outlet guide vane assembly includes a first actuation assembly arranged radially outward of the aft fixed portion and including a cam rod with two cams attached thereto and associated with the first tip and hub segments, rotation of the cam rod causing rotation of the cams which causes rotation of the first tip and hub segments;
- (4) FIG. 4A is a top cross-sectional view of either of the first tip and hub segments of the outlet guide vane of FIG. 3, showing a plurality of outlet guide vanes arranged circumferentially adjacent to each other, and showing that each outlet guide vane can include an actuation assembly having a respective cam rod and cams attached thereto, each configured to rotate a respective first tip or hub segment, and showing that each cam rod can be coupled to an annular ring;
- (5) FIG. 4B is a top cross-sectional view of either of the first tip and hub segments of the outlet guide vane of FIG. 3, showing different rotational positions of the first tip or hub segment based on rotational positions of the cam rod;
- (6) FIG. 5A is a side cutaway perspective view of a plurality of the first variable leading edge outlet guide vanes of FIG. 3, showing that the first actuation heads are connected to a full annular ring via actuation arms;
- (7) FIG. 5B is a side cutaway perspective view of a plurality of the first variable leading edge outlet guide vanes of FIG. 3, showing that two groups of first actuation heads are connected to annular ring segments via actuation arms, the segments being spaced apart circumferentially;
- (8) FIG. 5C is a side cutaway perspective view of a plurality of the first variable leading edge outlet

guide vanes of FIG. 3, showing that the first actuation heads are each connected to individual actuators, which are connected to a control system;

(9) FIG. 6 is a perspective view of the cam rod and cams of the outlet guide vane of FIG. 3;

(10) FIG. 7 is a side cross-sectional view of the outlet guide vane of FIG. 3, showing that the guide vane can further include a first air manipulating member arranged radially between the first tip segment and the first hub segment;

(11) FIG. 8 is a top cross-sectional view of the outlet guide vane of FIG. 7, showing the first tip segment at a neutral position and a top view of the first air manipulating member configured as a winglet;

(12) FIG. 9 is a perspective view of the outlet guide vane of FIG. 8, showing the first air manipulating member configured as a winglet;

(13) FIG. 10 is a perspective view of the outlet guide vane of FIG. 8, showing the first air manipulating member configured as a winglet;

(14) FIG. 11 is a top cross-sectional view of the outlet guide vane of FIG. 8, showing a top view of the first air manipulating member configured as a seal;

(15) FIG. 12 is a side cross-sectional view of a first variable leading edge outlet guide vane according to a further aspect of the present disclosure, showing that the first variable leading edge outlet guide vane includes the first tip segment and the first hub segment, showing that the outlet guide vane assembly includes a first actuation assembly arranged radially outward of the aft fixed portion and including a cam rod with two cams attached thereto and associated with the first tip and hub segments, rotation of the cam rod causing rotation of the cams which causes rotation of the first tip and hub segments, and showing that the guide vane further includes a static central portion arranged radially between the first tip segment and the first hub segment;

(16) FIG. 13 is a perspective view of the outlet guide vane of FIG. 12, showing that air manipulating members configured as a winglets can be arranged between the first tip and hub segments and the static central portion;

(17) FIG. 14 is a side cross-sectional view of a first variable leading edge outlet guide vane according to a further aspect of the present disclosure, showing that the first variable leading edge outlet guide vane includes first and second tip segments and first and second hub segments, showing that the outlet guide vane assembly includes a first actuation assembly arranged radially outward of the aft fixed portion and including a cam rod with four cams attached thereto and associated with the first and second tip and hub segments, rotation of the cam rod causing rotation of the cams which causes rotation of the respective tip and hub segments, and showing that the guide vane further includes a static central portion arranged radially between the first tip segment and the first hub segment;

(18) FIG. 15 is a perspective view of the outlet guide vane of FIG. 14, showing that air manipulating members configured as a winglets can be arranged between the first and second tip and hub segments and the static central portion;

(19) FIG. 16 is a side cross-sectional view of a first variable leading edge outlet guide vane according to a further aspect of the present disclosure, showing that the first variable leading edge outlet guide vane includes first and second tip segments and first and second hub segments, showing that the outlet guide vane assembly includes a first actuation assembly arranged radially outward of the aft fixed portion and including a cam rod with two cams attached thereto and associated with the first and second tip segments, showing that the outlet guide vane assembly includes a second actuation assembly arranged radially inward of the leading edge portion and including a cam rod with two cams attached thereto and associated with the first and second hub segments, and showing that the guide vane further includes a static central portion arranged radially between the first tip segment and the first hub segment;

(20) FIG. 17 is a side cross-sectional view of a first variable leading edge outlet guide vane according to a further aspect of the present disclosure, showing that the first variable leading edge

outlet guide vane includes first tip and hub segments, showing that the outlet guide vane assembly includes a first actuation assembly arranged radially outward of the aft fixed portion and including a cam rod with a cam attached thereto and associated with the first tip segment, and showing that the outlet guide vane assembly includes a second actuation assembly arranged radially inward of the leading edge portion and including a cam rod with a cam attached thereto and associated with the first hub segment;

(21) FIG. **18** is a side cross-sectional view of a first variable leading edge outlet guide vane according to a further aspect of the present disclosure, showing that the first variable leading edge outlet guide vane includes first tip and hub segments, showing that the outlet guide vane assembly includes a first actuation assembly arranged radially outward of the aft fixed portion and including a cam rod with a cam attached thereto and associated with the first tip segment, and showing that the outlet guide vane assembly includes a second actuation assembly arranged radially inward of the leading edge portion and including a cam rod with a cam attached thereto and associated with the first hub segment, and showing that the guide vane further includes a static central portion arranged radially between the first tip segment and the first hub segment; and

(22) FIG. **19** is a top cross-sectional view of a first variable leading edge outlet guide vane according to a further aspect of the present disclosure, showing that the cams of the cam rod directly engage with a tab formed on the axially aft side of the first tip or hub segment so as to rotate the first tip or hub segment, and showing different rotational positions of the first tip or hub segment based on rotational positions of the cam.

DETAILED DESCRIPTION

(23) For the purposes of promoting an understanding of the principles of the disclosure, reference will now be made to a number of illustrative embodiments illustrated in the drawings and specific language will be used to describe the same.

(24) An illustrative aerospace gas turbine engine **10** includes a fan assembly **12** and an engine core **13** having a compressor **14**, a combustor **16**, and a turbine **18**, as shown in FIG. **1**. The fan assembly **12** is driven by the turbine **18** and provides thrust for propelling an air vehicle by forcing fan exit air **15** through a fan duct **20** that circumferentially surrounds an outer casing **17** of the engine core **13**. The compressor **14** compresses and delivers air to the combustor **16**. The combustor **16** mixes fuel with the compressed air received from the compressor **14** and ignites the fuel. The hot, high-pressure products of the combustion reaction in the combustor **16** are directed into the turbine **18** to cause the turbine **18** to rotate about a central axis **11** and drive the compressor **14** and the fan **12**. In some embodiments, the fan may be replaced with a propeller, drive shaft, or other suitable configuration.

(25) The fan assembly **12** includes a fan **21** having a plurality of fan blades **22** that extend radially outward relative to the central axis **11** and that are located in the inlet of the gas turbine engine **10**, as shown in FIGS. **1** and **2**. The fan blades **22** direct at least a portion of the air flowing over the blades **22**, this portion being fan exit air **15** as shown in FIGS. **1** and **2**, through the fan duct **20** such that the fan exit air **15** bypasses the engine core **13** and provides additional thrust for the gas turbine engine **10**. The fan duct **20** includes an outer fan duct casing **19** and an inner wall **23** that together define an annular fan duct passage **24** through which the fan exit air **15** flows and subsequently exits the fan duct **20** into the ambient air surrounding the engine **10**. The inner wall **23** may include an axially forward end **23S** that functions as a splitter such that a portion of the incoming fan exit air **15** enters the engine core through the engine core passage **23P** radially inward of the inner wall **23** and a portion of the fan exit air **15** enters the fan duct **20**.

(26) In the illustrative embodiment, the fan assembly **12** further includes outlet guide vane assembly **28** located in the fan duct **20** axially downstream of the plurality of fan blades **22** that is configured to reduce the incidence between the outlet guide vane assembly **28** and the fan exit air **15** received from the plurality of fan blades **22** and return the flow to generally axial flow, as shown in FIG. **2**. In some embodiments, the outlet guide vane assembly **28** is arranged axially downstream

of the axially forward end **23S** of the inner wall **23**. In the illustrative embodiment, the outlet guide vane assembly **28** includes a first variable leading edge outlet guide vane **30** having a leading edge portion **31** and a fixed aft portion **50**, as shown in FIG. **3**. The leading edge portion **31** includes a first tip segment **32** configured to rotate about a leading edge pitch axis **39** and a first hub segment **42** located radially inward of and separate from the first tip segment **32**, also configured to rotate about the leading edge pitch axis **39**.

(27) The first variable leading edge outlet guide vane **30** extends radially outward relative to the central axis **11**, as shown in FIG. **3**. In some embodiments, the first variable leading edge outlet guide vanes **30** may include a plurality of first variable leading edge outlet guide vanes **30** disposed around a circumferential extent of the inner wall **23** define a first outlet guide vane stage.

(28) As shown in FIG. **3** and contextualized in FIG. **4A** and FIG. **4B**, the first variable leading edge outlet guide vane **30** includes an airfoil shape. The first tip segment **32** includes a leading edge **33** located at a forward end of the segment **32**, a trailing edge **34** axially spaced apart from the leading edge **33** and located at an aft end of the segment **32**, a radially outer side **35**, and a radially inner side **36**. Similarly, the first hub segment **42** includes a leading edge **43** located at a forward end of the segment **42**, a trailing edge **44** axially spaced apart from the leading edge **43** and located at an aft end of the segment **42**, a radially outer side **45**, and a radially inner side **46**.

(29) Together, the first tip and hub segments **32**, **42** form the leading edge portion **31**. As shown in FIG. **4A**, the leading edge portion **31** and the fixed aft portion **50** together form the overall vane **30**, which includes a pressure side surface **31P** that extends between the leading edges **33**, **43** of the segments **32**, **42** and a trailing edge **52** of the fixed aft portion **50** on one side of the vane **30**.

Similarly, a suction side surface **31S** extends between the leading edges **33**, **43** of the segments **32**, **42** and a trailing edge **52** of the fixed aft portion **50** on an opposing side of the vane **30**.

Illustratively, the leading edge portion **31** is approximately one-third of the total chord length **30H** of the vane **30**, while the fixed aft portion **50** is approximately two-thirds of the total chord length **30H**. In other embodiments, the leading edge portion **31** is approximately one-fourth of the total chord length **30H** of the vane **30**, while the fixed aft portion **50** is approximately three-fourths of the total chord length **30H**. In other embodiments, the leading edge and fixed aft portions **31**, **50** are each half the total chord length **30H**.

(30) The first tip and hub segments **32**, **42** are configured to rotate about the leading edge pitch axis **39**, as shown in FIG. **3**. In the illustrative embodiment, the leading edge pitch axis **39** is located closer to the trailing edges **34**, **44** of the segments **32**, **42** than the leading edges **33**, **43**. In this manner, the first tip and hub segments **32**, **42** can rotate proximate to the fixed aft portion **50** such that airflow remains uninterrupted while flowing over the vane **30**. In some embodiments, a hub member **19H** of the first tip segment **32** is rotatably coupled to the outer fan duct casing **19** within a hub receiving recess **19R**. Similarly a hub member **23H** of the first hub segment **42** is rotatably coupled to the inner wall **23** within a hub receiving recess **23R**.

(31) The first hub segment **42** is configured to independently rotate about the leading edge pitch axis **39** relative to the first tip segment **32**, and may include a small radial gap therebetween. In some embodiments, the first hub segment **42** can include a hinge rod **60** extending radially outwardly from the radially outer side **45** of the first hub segment **42** and rotatably received within the radially inner side **36** of the first tip segment **32** to add stability to the rotation of the two segments **32**, **42** relative to each other while allowing for independent rotation. The hinge rod **60** of the first hub segment **42** can be cylindrical and aligned with the leading edge pitch axis **39**. In some embodiments in which the first tip and hub segments **32**, **42** are formed of a sufficiently stiff material, a hinge rod may be omitted.

(32) To rotate independently, the illustrative embodiment of the outlet guide vane assembly **28** includes a first actuation assembly **70** associated with the first tip and hub segments **32**, **42**. Illustratively, the first actuation assembly **70** includes a first actuation arm **74**, a first actuation head **76**, and a cam rod **78** fixedly couple to the first actuation head **76** and extending radially. The first

actuation assembly **70** further includes a first cam **54** and a second cam **57** selectively coupled to the cam rod **78** and spaced apart from each other in the radial direction along the length of the cam rod **78**. The cams **54**, **57** engage cam actuation rods **55**, **58** that are rotatably coupled to the first tip and hub segments **32**, **42**, respectively, which in turn rotates the first tip and hub segments **32**, **42**. As will be described in detail below, the cams **54**, **57** may be selectively clocked along the cam rod **78** so as to preselect pitch angles to which rotation of the cams **54**, **57** will rotate the first tip and hub segments **32**, **42**.

(33) Illustratively, the first actuation arm **74** and the first actuation head **76** are arranged radially outward of the fixed aft portion **50**, as shown in FIG. **3**. Although the first actuation head **76** is shown in this manner, and the cam rod **78** is shown as extending radially inwardly away from the first actuation head **76**, one of ordinary skill in the art would understand that the actuation assembly components could be arranged oppositely, in particular having the first actuation head **76** arranged radially inward of the fixed aft portion **50**, and thus having the cam rod **78** extending radially outwardly away from the first actuation head **76**.

(34) In some embodiments, the first actuation arm **74** extending axially away from the first actuation head **76**, as shown in FIG. **3**. In some embodiments, the first actuation head **76** extends into and is supported by the outer fan duct casing **19** and is fixedly coupled to the cam rod **78** which extends radially therefrom. As shown in FIG. **3**, the first actuation head **76** is axially offset in the aft direction from the leading edge pitch axis **39**. In some embodiments, the cam rod **78** is arranged closer to the forward side **53** of the fixed aft portion **50** than the trailing edge **52**. In particular, the cam rod **78** is located approximately halfway along the chord length **30H** of the vane **30**, or slightly offset therefrom. For example, the cam rod **78**, which rotates the cams **54**, **57** about their pivot axes, is located slightly axially aft of the halfway point **30H2** of the chord length **30H**.

(35) In some embodiments, the actuation head **76** is cylindrical having a central axis **76C** that aligns with a central axis **78C** of the cam rod **78**. The first actuation head **76** is fixedly arranged within an opening **77** formed at a first end **74A** of the first actuation arm **74**. As a result, pivoting movement of the first actuation arm **74** about the actuation head **76** central axis **76C** rotates the first actuation head **76**, which in turn rotates the cam rod **78**. In some embodiments, the first actuation arm **74** is arranged radially outward of the vane **30** and the outer fan duct casing **19**, as shown in FIG. **3**.

(36) As shown in FIG. **4A** and FIG. **4B**, the cam rod **78** is configured to rotate about its central axis **78C**, which in turn will rotate the cams **54**, **57**. Each cam **54**, **57** includes a non-circular shape, or a cam shape, having an axially forward facing surface **54C**, **57C** that is configured to engage an axially aft terminal end of the respective cam actuation rod **55**, **58**. As shown in FIG. **4A** and FIG. **4B**, the cam rod **78** may be offset from the fixed aft portion **50** and extend along a side of the fixed aft portion **50**, but in other embodiments, as shown in FIG. **3**, a cam rod receiving cavity **51** is formed in the fixed aft portion **50** through which the cam rod **78** may extend.

(37) In some embodiments, the cavity **51** include enlarged cavity portions **51A**, **51B** that contain the cams **54**, **57** therein, as shown in FIG. **3** and FIG. **4A**. The enlarged cavity portions **51A**, **51B** are sized so as to contain the cams **54**, **57** therein without the cams **54**, **57** protruding outwardly beyond the walls of the vane **30**. As such, the cams **54**, **57** are prevented from interrupting any portion of the air flow over the vane **30**. The enlarged cavity portions **51A**, **51B** may be entirely sealed off from the environment via the pressure and suction side **31P**, **31S** walls, except for where the actuation rods **55**, **58** must extend axially forward out of the fixed aft portion **50** and engage the tip and hub segments **32**, **42**.

(38) The first cam actuation rod **55** is rotatably coupled to the first tip segment **32** and extends axially aft therefrom toward the first cam **54**, as shown in FIGS. **3-4A**. Similarly, the second cam actuation rod **58** is rotatably coupled to the first hub segment **42** and extends axially aft therefrom toward the second cam **57**. The first cam actuation rod **55** is rotatably coupled to the first tip segment **32** at a point offset from the leading edge pitch axis **39** (centrally located relative to **19H**

as shown in FIGS. 4A and 4B) in the circumferential direction such that axially forward movement of the first cam actuation rod 55 rotates the first tip segment 32 in the first rotational direction (downward direction when viewing FIGS. 4A and 4B). Similarly, the second cam actuation rod 58 is rotatably coupled to the first hub segment 42 at a point offset from the leading edge pitch axis 39 in the circumferential direction such that axially forward movement of the second cam actuation rod 58 rotates the first hub segment 42 in the first rotational direction. In some embodiments, the connection to the first tip and hub segments 32, 42 is sufficient to support the actuation rods 55, 58, but in other embodiments, the actuation rods 55, 58 may be supported on the fixed aft portion 50 or any other means known in the art.

(39) Illustratively, the first tip segment 32 includes a first cam actuation rod receiving recess 37 formed at least partially in the aft side 34 of the first tip segment 32, as shown in FIG. 3. The first cam actuation rod 55 is rotatably mounted within the first cam actuation rod receiving recess 37. In some embodiments, the forward end of the first cam actuation rod 55 includes radially outer and inner pivot pins 56A, 56B that can be inserted into corresponding holes formed in the recess 37. Similarly, the first hub segment 42 includes a second cam actuation rod receiving recess 47 formed at least partially in the aft side 44 of the first hub segment 42. The second cam actuation rod 58 is rotatably mounted within the second cam actuation rod receiving recess 47. In some embodiments, the forward end of the second cam actuation rod 58 includes radially outer and inner pivot pins 59A, 59B that can be inserted into corresponding holes formed in the recess 47.

(40) In order to rotate the first tip segment 32, an axially aft end of the first cam actuation rod 55 engages the axially forward facing surface 54C of the first cam 54 such that rotation of the first cam 54 in the first rotational direction causes the axially forward facing surface 54C to move the first cam actuation rod 55 axially forward. FIG. 4B shows various rotational positions of the cams 54, 57 and the first tip and hub segments 32, 42. When the first cam actuation rod 55 moves axially forward, the first tip segment 32 is rotated in the first rotation direction. As can be seen in FIG. 4B, the more the cam 54, 57 is rotated, the more the first tip and hub segments 32, 42 rotate.

(41) Similarly, an axially aft end of the second cam actuation rod 58 engages the axially forward facing surface 57C of the second cam 57 such that rotation of the second cam 57 in the first rotational direction causes the axially forward facing surface 57C to move the second cam actuation rod 58 axially forward. When the second cam actuation rod 58 moves axially forward, the first hub segment 42 is rotated in the first rotation direction. In some embodiments, one or both of the actuation rods 55, 58 can include a spring 55S, 58S that returns the actuation rods 55, 58 and thus the first tip and hub segments 32, 42 back in the axially aft direction when the cams 54, 57 are rotated in the opposing, second rotation direction.

(42) The cams 54, 57 can be selectively clocked along the cam rod 78 such that their rotation can cause the first tip and hub segments 32, 42 to be rotated to selectable first and second pitch angles, respectively. For example, as can be seen in FIG. 6, the first cam 54 can include a first cam hole 54H through which the cam rod 78 extends, and the second cam 57 can include a second cam hole 57H through which the cam rod 78 extends. The cams 54, 57 can be selectively clocked into place, such as, for example, the position of the first cam 54 and the position of the second cam 57 shown in FIG. 6. Due to the difference in the clocked positions, the first tip and hub segments 32, 42 will move to different pitch angles due to the rotation of the cams 54, 57.

(43) In this way, the first tip and hub segments 32, 42 can be individually controlled with respect to each other to the same or differing angles, thus allowing for a multitude of segment positions and arrangements to be achieved. This provides for great flexibility in managing incoming airflows which may include distortions and disturbances. This can be particularly useful in embedded engine applications with complex intake and inlet duct geometries. In such scenarios, the distortion flows have more significant gradients and vortices, even in flight, so accommodation is necessary to maintain fan operability and performance.

(44) As will be described in greater detail below, the first actuation arm 74 is configured to be

moved generally circumferentially relative to the first actuation head **76** by annular rings (such as annular ring **62** shown in FIG. **3** and FIG. **4A**), segmented annular rings, or individual actuators so as to pivot the actuation arm **74** about the actuation head **76** central axis **76C**. In particular, the actuation arm **74** may be selectively pivoted about the actuation head **76** central axis **76C** so as to rotate the cam rod **78** which rotates the first and second cams **54**, **57** which rotates the first tip and hub segments **32**, **42**. As a result, the first tip segment **32** can be selectively rotated about the leading edge pitch axis **39** to a first pitch angle relative to the incoming fan exit air **15**. Similarly, the first hub segment **42** can be selectively rotated about the leading edge pitch axis **39** to a second pitch angle relative to the incoming fan exit air **15**. The fixed trailing edge **50** rotates any flow back to near axial to minimize loss from swirl.

(45) As shown in FIG. **3**, and in greater detail in FIGS. **5A**, **5B**, **5C**, the outlet guide vane assembly **28** can further include a plurality of variable leading edge outlet guide vanes **30** arranged around the entirety of the circumferential extent of the vane assembly **28**. The plurality of variable leading edge outlet guide vanes **30** include the first variable leading edge outlet guide vane **30** described above. Each variable leading edge outlet guide vane **30** of the plurality of variable leading edge outlet guide vanes **30** includes a respective tip segment **32** and hub segment **42**, as well as the actuation assembly **70**. The first actuation arms **74** of each vane **30** may be coupled to an annular rings (for example, first annular ring **62**) that extends circumferentially about the central axis **11**. The actuation arm **74** may be coupled to the ring **62** via an opening formed at a second end **74B** of the actuation arm **74**.

(46) As can be seen in FIG. **5A**, the outlet guide vane assembly **28** may include a fully annular first annular ring **62**. In this configuration, the first annular rings **62** can be configured to rotate about the central axis **11** so as to move the actuation arm **74** coupled thereto generally circumferentially relative to the first actuation head **76**, or in other words, to pivot the actuation arm **74** coupled thereto about the actuation head **76** central axis **76C**. As a result, the rotation of the first annular ring **62** about the central axis **11** causes the cam rod **78** to rotate, thus causing the cams **54**, **57** to rotate, thus causing rotation of the first tip and hub segments **32**, **42**, respectively. As will be described in detail below, the annular ring **62** may be controlled by a control system **99**.

(47) In some scenarios, it may be beneficial to have more control over individual sections of the plurality of variable leading edge outlet guide vanes **30** or individual vanes **30**. This may be particular useful when the engine experiences more significantly distorted flows, such as in embedded engine applications. Sectional control or individual control would provide additional flexibility to accommodate as much flow variation as possible to recover operability and performance margins. For example, as an aircraft maneuvers through ground crosswind or variations of flight orientations such as sideslip and pitch variations, the plurality of variable leading edge outlet guide vanes **30** can be rotationally moved in sections or independently to best reset the inlet angles and maximize the fan operability envelope. The optimized rest of the inlet angles improves stall margin, improves efficiency and performance, and reduces fan forcing debits.

(48) As can be seen in FIG. **5B**, first annular ring **62** may be broken up into segments so as to include multiple circumferential segments disposed about the central axis **11**. In particular, the multiple circumferential segments are radially aligned with each other and circumferentially spaced apart. For example, the assembly **28** may include a first annular ring segment **62S1** and a second annular ring segment **62S2** radially aligned with the first annular ring segment **62S1**, extending partially circumferentially about the central axis **11**, and circumferentially offset from the first annular ring segment **62S1**.

(49) In this configuration, the first and second annular ring segments **62S1**, **62S2** can be configured to independently rotate about the central axis **11** so as to move each actuation arm **74** coupled thereto generally circumferentially relative to the first actuation head **76**, or in other words, to pivot each actuation arm **74** coupled thereto about the actuation head **76** central axis **76C**. The first and second annular ring segments **62S1**, **62S2** are circumferentially spaced apart a great enough

distance to allow maximum movement relative to each other. For example, in some scenarios, the first annular ring segment **62S1** may be moved in an opposing circumferential direction to the second annular ring segment **62S2** such that the respective tip segments **32** of their associated vanes **30** are rotated in opposing rotational directions.

(50) The outlet guide vane assembly **28** can include multiple annular ring segments similar to the segments **62S1**, **62S2** disposed around the entire circumferential extent of the assembly **28**. As such, the multiple annular ring segments can define groups of vanes **30**, the tip and hub segments **32**, **42** of each being able to be rotated to unique pitch angles based on the circumferential movement and position of their respective annular ring segment. As will be described in detail below, the annular ring segments **62** may be controlled by a control system **99**.

(51) In some embodiments, the control system **99** is configured to rotate each segment **32**, **42** of each vane **30** of the first plurality of variable leading edge outlet guide vanes **30** individually relative to each other vane's **30** segments **32**, **42**, as shown in FIG. **6**. That is to say, each segment **32**, **42** of each vane **30** may be rotated without moving any of the other vanes' segments **32**, **42** of the first and second plurality of variable leading edge outlet guide vanes **30**. This allows for the vanes **30** to be controlled in a variety of configurations. For example, a group of the plurality of variable leading edge outlet guide vanes **30** may be controlled to be rotated in unison and rotated to a first pitch angle, while other vanes **30** are individually controlled, each to unique pitch angles. The individual controllability of the vanes **30** accounts for variations in the fan exit air **15** around the circumference of the area of the plurality of variable leading edge outlet guide vanes **30**, which is particularly beneficial for the reasons discussed above.

(52) In another example, if the rotation of the first plurality of guide vanes **30** causes more undesirable flow effects in certain circumferential sectors, only vanes **30** located in those certain circumferential sectors may be rotated to specific pitch angles to reduce losses from said flow effects, while other vanes **30** may be rotated to different pitch angles. Then, if the flow dynamically encounters different distortions during operation, different circumferential sectors may require vane **30** adjustment. The individual controllability allows for the option to adjust these different circumferential sectors dynamically.

(53) In order to carry out the individual control, the second end **74B** of the actuation arm **74** may include a control member **75** coupled to an external actuator (not shown) that moves the actuation arm **74** generally circumferentially relative to the first actuation head **76**, as shown in FIG. **5C**. In other embodiments, the first actuation head **76** may be individually actuated via an actuator coupled directly to the actuation head **76**, for example an actuator coupled to a radially outer side of the first actuation head **76**.

(54) As touched on above, by controlling at least one vane **30** of the plurality of variable leading edge outlet guide vanes **30**, the control system **99** is configured to control at least some of the flow of the fan exit air **15** after it passes over and exits the fan blades **22**. By controlling the entirety of the plurality of variable leading edge outlet guide vanes **30**, the control system **99** can accommodate the overall flow of the fan exit air **15**, in particular distorted flow, after it passes over and exits the fan blades **22** in order to control fan blade **22** response to forces acting on the fan blades **22**, as well as to reduce losses created by undesirable variations in the air flow. Moreover, because the fan exit air **15** may not be uniform as it exits the fan blades **22**, the plurality of variable leading edge outlet guide vanes **30** or the axial passage between the vanes **30** and the fan blades **22** operate further from their ideal design conditions. By adjusting the plurality of variable leading edge outlet guide vanes **30**, parameters such as incidence are improved, and detrimental flow conditions and losses in the vanes **30** or the axial passage between the vanes **30** and the fan blades **22** such as vortices and stall are reduced.

(55) In some embodiments, the control system **99** is configured to rotate the segments **32**, **42** of each vane **30** of the first plurality of variable leading edge outlet guide vanes **30** to a first vane-pitch angle in response to the gas turbine engine **10** operating at a given operating condition so as

to reduce the incidence between the outlet guide vane assembly **28** and the fan exit air **15** received from the plurality of fan blades **22** and redirect the fan exit air **15** in a first direction, in particular a generally axial direction. In particular, the operating condition in which the fan assembly **12** and gas turbine engine **10** are operating in may include at least one of take-off, climb, cruise, landing, and aircraft maneuvers of an aircraft having the engine **10** equipped. In each of these operating conditions, the plurality of fan blades **22** and/or the vanes **30** of the fan assembly **12** may experience various undesirable operability issues such as forcing, stall, and flutter. For example, the engine **10** may operate in particular speed ranges for each of the operating conditions, and as result, the fan blades **22** may experience greater or lower levels of forcing, stall, and/or flutter in response to the engine **10** operating in particular speed ranges.

(56) In order to compensate for these forces acting on the fan blades **22**, the control system **99** is configured to rotate the segments **32**, **42** of the first plurality of variable leading edge outlet guide vanes **30** to an arrangement of first vane-pitch angles in order to alter the angle of the flow of fan exit air **15** after it exits the fan blades **22**. This change in the angle of flow as the fan exit air **15** passes over the first plurality of variable leading edge outlet guide vanes **30** reduces the amount of forcing, stall, and/or flutter experienced by the fan blades **22** and/or the outlet guide vanes **30**. Moreover, the control system **99** is configured to reset a desired incidence of air flow into the first plurality of variable leading edge outlet guide vanes **30** in response to swirl in the inlet flow. This produces an averaging effect that improves engine performance and efficiency. These arrangements of the vanes **30** can also recover the losses created by flow separation, flow distortions, vortices, and/or swirl.

(57) The control system **99** is operable to control the segments **32**, **42** of the first plurality of variable leading edge outlet guide vanes **30** in a variety of configurations and arrangements in order to compensate for inlet pressure distortion, vortices and swirl, thus reducing the forcing, stall, flutter, flow separation, and any other undesirable effects in the fan rotor or outlet vanes. For example, in some embodiments, the control system **99** is configured to rotate the segments **32**, **42** of each vane **30** of the first plurality of guide vanes **30** in unison. In other words, all of the first plurality of guide vanes **30** move to the same first vane-pitch angle. In such embodiments, each vane **30** may be mechanically connected to each other via the first and second annular rings **62**, **64**.

(58) In some embodiments, the control system **99** is configured to rotate at least two different groups of variable leading edge outlet guide vanes **30**. For example, the control system **99** may be configured to selectively rotate the segments **32**, **42** of each group of vanes **30** to create non-uniform backpressure that drives the fan inlet distortion flows within the fan to change or redistribute around the circumference of the fan. This locally reduces loading on fan blades **22** within a lip separated flow with low local pressure to reduce forcing and/or improve the uniformity of flow in general through the fan to reduce forcing. In particular, fully opening (allowing full flow through the guide vanes) at least one group of vanes **30** and fully closing at least one further group of vanes **30** (allowing no flow through the guide vanes) reduces a tendency for a local stall of the fan blades **22** that could lead to early overall stall in the fan. In some embodiments, the control system **99** is configured to rotate a large group of vanes **30** which counters bulk swirling flows or local changes to improve localized intake swirl gradients to improve fan performance and operability.

(59) In at least one additional embodiment, the segments **32**, **42** of the plurality of variable leading edge outlet guide vane **30** are broken into unique groups of vanes **30**, as shown in FIG. 5B. Each group of vanes **30** is mechanically connected to each other, or ganged, via a unique circumferentially extending annular ring segment, such as the annular ring segments **62S1**, **62S2** shown in FIG. 5B. Although the illustrative embodiment shows each group of vanes **30** including two vanes **30**, the vanes **30** may be grouped and ganged in any combination of at least two groups of vanes totaling at least one fewer vane than the total number of vanes **30** in the plurality of outlet guide vanes **30**. For example, if the first plurality of outlet guide vanes **30** includes 60 vanes, a first

group may include 30 vanes and a second group may include 30 vanes. As a further non-limiting example, a first group of vanes may include 50 vanes, a second group of vanes may include five vanes, and a third group of vanes may include five vanes.

(60) In some embodiments, the segments **32, 42** of the first plurality of variable leading edge outlet guide vanes **30** includes a first group of first vanes **30** and a second group of the segments **32, 42** of first vanes **30** different from the first group of guide vanes **30**. The control system **99** is configured to rotate the first group of first vanes **30** to a first vane-pitch angle and the second group of first vanes **30** to a second vane-pitch angle that is different from the first vane-pitch angle. The groups of vanes **30** may be individually controlled or each group may be ganged together. For example, in some embodiments, one half of the first plurality of outlet guide vanes **30** is the first group and the other half of the first plurality of outlet guide vanes **30** is the second group.

(61) In some embodiments, the control system **99** utilizes predetermined arrangements of the plurality of variable leading edge outlet guide vanes **30** that are based on predetermined measurements and data taken in predetermined engine operating conditions and predetermined airflow characteristics. As such, the control system **99** is configured to rotate the vanes **30** to specific predetermined arrangements based on the operating condition and/or airflow characteristic(s) of the fan exit air **15** or the inlet air that the engine **10** and fan assembly **12** are operating in, or based on projected operating conditions and/or airflow characteristic(s) that will be encountered by the engine **10** during a mission.

(62) In some embodiments, the predetermined arrangements of the segments **32, 42** of the plurality of variable leading edge outlet guide vanes **30** can be based on previously acquired test data corresponding to specific flight conditions. In a more complex arrangement, the control system **99** could be coupled to measurement systems, such as the at least one sensor **92** described below, that detect flow instabilities associated with impending fan stall to direct vane geometry changes to extend margins. The control system **99** may be more effective near stall as the vanes **30** can redistribute flow conditions to minimize local stall cells. Having smaller ganged vane **30** groups are efficient as well in that such groups only reduce losses locally to extend operability and performance.

(63) In some embodiments, the control system **99** includes at least one sensor **92** configured to take real-time measurements of the air flow within the fan duct passage **24** and of forces acting on the fan assembly components, as shown in FIG. 2. The real-time measurements may be utilized in order to determine the operating condition and/or airflow characteristic(s) of the fan exit air **15** or the inlet air that the engine **10** and fan assembly **12** are operating in so as to inform the control system **99** to which predetermined arrangement to rotate the plurality of variable leading edge outlet guide vanes **30**.

(64) In some embodiments, the control system **99** includes a neural network configured to perform machine learning such that the control system **99** can iterate over the predetermined arrangements in order to calculate new arrangements that are applicable to new variations in the operating condition and/or airflow characteristics that are unaccounted for by the predetermined settings and arrangements.

(65) In some embodiments, the control system **99** further includes a subsystem control that is integrated with other engine controls to further control reduction of losses created by undesirable variations in the air flow and improve engine performance and efficiency. For example, if rotation of vanes **30** resulted in a fan flow drop, the subsystem control is configured to compensate for this by increasing the fan speed in order to maintain thrust, and/or by changing the exhaust area of the engine **10** in order to further reduce the losses and improve engine efficiency.

(66) In some embodiments, the at least one sensor **92** may be located proximate to the fan blades **22**, proximate to the plurality of variable leading edge outlet guide vanes **30**, or both, as shown in FIG. 2. In some embodiments, the sensor or sensors **92** are located in the outer casing **19** radially outward of the fan blades **22** and vanes **30**. The at least one sensor **92** may include one of or a

combination of dynamic sensors, static wall pressure sensors, altitude sensors, sensors configured to detect the angle of attack of the plurality of fan blades **22**, sensors configured to detect the tip timing of the plurality of fan blades **22**, and airspeed sensors. In some embodiments, the sensor **92** may be a dynamic pressure transducer. The sensor **92** may also be a sensor configured to measure a rotational speed of the fan blades **22**, which could be used along with an additional sensor that is a dynamic pressure transducer.

(67) In the illustrative embodiment, the functionality of the control system **99** described herein may be implemented in various processing and computing devices, and may be located within the engine **10** or outside of the engine **10**. Moreover, the functionality may be configured to operate on executable software provided on the processing and computing devices.

(68) Furthermore, the functionality disclosed herein may be implemented in various configurations using computer-readable media for carrying or having computer-executable instructions or data structures stored thereon. Such computer-readable media can be any available media that can be accessed by a general purpose or special purpose computer. By way of example, and not limitation, such computer-readable media can comprise physical storage and/or memory media such as RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code means in the form of computer-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer. Computer-executable instructions comprise, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing device to perform a certain function or group of functions.

(69) As shown in FIG. 7, an air manipulating member **90** may be arranged in the small radial gap between the first tip and hub segments **32**, **42**. In particular, in some embodiments, the air manipulating member **90** may be configured as a winglet **90** or platform coupled to an axially forward side **53** of the fixed aft portion **50**. In other embodiments, the air manipulating member **90** may be configured as a seal **90S** coupled to the axially forward side **53** of the fixed aft portion **50**, or may be directly coupled to one of the sides **36**, **45** of the tip and hub segments **32**, **42**.

(70) As shown in FIGS. 8-10, the air manipulating member **90** may be configured as a winglet **90** that reduce radial flows across the tip and hub segments **32**, **42**. This is particularly helpful when the tip and hub segments **32**, **42** are set to non-neutral, opposing pitch angles. As shown in FIG. 8, the winglet **90** may be generally planar and include a radially outer winglet surface **91** and a radially inner winglet surface **92**, each surface generally facing radially. The winglet **90** further includes an axially aft end **93**, a suction side edge **94**, and a pressure side edge **95**. The suction and pressure side edges **94**, **95** may be curved similarly to the curvature of the airfoil shape of the vane **30**.

(71) The winglet **90** further includes a forward edge **96** that is curved. In some embodiments, the forward edge **96** is curved to match the rotational path, or path of movement, of the leading edge **33**, **43** of the tip and hub segments **32**, **42**. In particular, the forward edge **96** may include a radius of curvature **97** that matches the rotational path of the tip and hub segments **32**, **42**, as shown in FIG. 8. In some embodiments, a width **90W** of the winglet **90** is equal to or greater than a width **30W** of the first tip segment **32** and the first hub segment **42** along an entire axial extent of the first tip segment **32** and the first hub segment **42**. In some embodiments, the width of the winglet **90** at the axially aft end **93** is less than a length of the forward edge **96**. In some embodiments, the winglet **90** has a radial height that allows for the seal to barely contact or nearly contact the sides **36**, **45** of the tip and hub segments **32**, **42** so as to prevent flow from entering this gap.

(72) FIG. 9 and FIG. 10 show a side and front perspective view, respectively, of the winglet **90**. In the example shown in FIG. 9, the tip segment **32** is moved to a first pitch angle, which is to the left of a neutral zero angle position when viewing FIG. 10 and FIG. 11. As can be seen, the forward edge **96** is long enough such that the pressure side edge **95** is located beyond the position of the tip

segment **32** in the circumferential direction. In some embodiments, the forward edge **96** is formed to be long enough such that both the suction and pressure side edges **94, 95** are located outside of the potential range of pitch angle positions of the tip and hub segments **32, 42**. This ensures that radial flows across the tip and hub segments **32, 42** will continue to be better matched at all pitch angle positions of the tip and hub segments **32, 42**.

(73) As can be seen in FIG. **11**, the air manipulating member **90** may be configured as a seal **90S** which is coupled to the axially forward side **53** of the fixed aft portion **50** at an axially aft end **93S** of the seal **90S**. The seal **90S** may be generally planar. In some embodiments, the seal **90S** has a radial height that allows for the seal to barely contact or nearly contact the sides **36, 45** of the tip and hub segments **32, 42** so as to seal the radial gap between the segments **32, 42** and prevent flow from entering this gap. In some embodiments, the seal **90S** may be directly coupled to the radially inner side **36** of the tip segment **32**. In some embodiments, the seal **90S** may be directly coupled to the radially outer side **45** of the hub segment **42**. In some embodiments, a width **90SW** of the seal **90S** is equal to or greater than a width **30W** of the first tip segment **32** and the first hub segment **42** along an entire axial extent of the first tip segment **32** and the first hub segment **42**. In some embodiments, the seal **90S** may include a cutout **98** so as to allow a hinge rod **37** to pass therethrough. The winglet **90** may include a similar cutout **98**. In other embodiments in which the segments **32, 42** do not include a hinge rod, the cutout **98** would not be necessary. In some embodiments, an outer perimeter of the seal **90S** generally corresponds to an outer perimeter of each of the first tip segment **32** and the first hub segment **42**, as shown in FIG. **11**.

(74) A person skilled in the art will understand that all features and components of all embodiments described herein, including the guide vane assembly **28** described above and the outlet guide vane assemblies described below can be interchanged and modified to include some elements of some embodiments and other elements of other embodiments. For example, even though some embodiments are shown without a static central segment, these embodiments may include such a central segment, as described herein. Similarly, although some embodiments described two tip segments and two hub segments, some embodiments can include one tip and two hub segments, two tip and one hub segments, each with or without a static central hub section, or other combinations thereof. Similarly, the actuation assemblies can be interchanged, with some embodiments and configurations including pass-through actuation assemblies, two assemblies arranged at the tip and hub, different configurations of segmented and full annular rings, and the like.

(75) Another embodiment of an outlet guide vane assembly **128** is shown in FIG. **12**. The outlet guide vane assembly **128** is similar to the outlet guide vane assembly **28** shown in FIGS. **1-11** and described herein. Accordingly, similar reference numbers in the **100** series indicate features that are common between the outlet guide vane assembly **128** and the outlet guide vane assembly **28**. The description of the outlet guide vane assembly **28** is incorporated by reference to apply to the outlet guide vane assembly **128**, except in instances when they conflict with the specific description and the drawings of the outlet guide vane assembly **128**.

(76) Similar to the outlet guide vane assembly **28** described above, the outlet guide vane assembly **128** includes first tip and hub segments **132, 142**. In this embodiment, the first variable leading edge outlet guide vane **130** further includes a central segment **148** arranged between the first tip segment **132** and the first hub segment **142** such that the first tip segment **132** and the first hub segment **142** are radially spaced apart. In some embodiments, the central segment may be coupled to and extend axially away from the axially forward side **153** of the fixed aft portion **150**. The central segment **148** is static and does not rotate.

(77) As can be seen in FIG. **12**, the central segment **148** includes a radially outer side **148A**, a radially inner side **148B**, and an axially forward end **148C** extending between the radially outer and inner sides **148A, 148B** and partially defining the leading edge of the vane **130** along with the tip and hub segments **132, 142**. The central segment **148** may further include a radially outer hinge rod

receiving cavity **148D** configured to receive a hinge rod **137A** of the tip segment **132**, and a radially inner hinge rod receiving cavity **148E** configured to receive a hinge rod **147A** of hub segment **142**. The central segment **148** may provide stability to the tip and hub segments **132**, **142**, as well as allow for uninterrupted flow over the central portion of the vane **130**. The first tip and hub segments **132**, **142** further include actuation rod receiving recesses **137B**, **147B** formed therein in that receive the actuation rods **155**, **158**, similar to the actuation rods **55**, **58** described above.

(78) In some embodiments, the outlet guide vane assembly **128** may include a single or multiple air manipulating members **190A**, **190B** arranged in the small radial gaps between the tip segment **132** and central segment **148** and between the hub segment **142** and central segment **148**. The air manipulating members **190A**, **190B** may be formed similarly to the air manipulating member **90** described above, in particular formed as seals or winglets.

(79) In some embodiments, one of the air manipulating members **190A**, **190B** may be formed as a winglet while the other is formed as a seal. In some embodiments, both of the air manipulating members **190A**, **190B** are formed as a seal. In some embodiments, both of the air manipulating members **190A**, **190B** are formed as a winglet. In embodiments in which the tip and hub segments **132**, **142** are rotationally attached to the central segment **148** via hinge rods similar to those described above, the air manipulating members **190A**, **190B** can include a cutout (not shown but similar to the cutout **98**) to allow the hinge rods to pass therethrough. In other arrangements in which the tip and hub segments **132**, **142** are entirely spaced apart from the central segment **148**, the cutout would not be necessary.

(80) As can be seen in FIG. **13**, the forward edge **196A** of the winglet **190A** is long enough such that the pressure side edge **195A** is located beyond the position of the tip segment **132** in the circumferential direction. Similarly, the forward edge **196B** of the winglet **190B** is long enough such that the suction side edge **194B** is located beyond the position of the hub segment **142** in the circumferential direction. In some embodiments, the forward edges **196A**, **196B** are formed to be long enough such that both the suction and pressure side edges **194A**, **194B**, **195A**, **195B** are located outside of the potential range of pitch angle positions of the tip and hub segments **132**, **142**. This ensures that radial flows across the tip and hub segments **132**, **142** will continue to be better matched at all pitch angle positions of the tip and hub segments **132**, **142**.

(81) Another embodiment of an outlet guide vane assembly **228** is shown in FIG. **14** and FIG. **15**. The outlet guide vane assembly **228** is similar to the outlet guide vane assemblies **28**, **128** shown in FIGS. **1-13** and described herein. Accordingly, similar reference numbers in the **200** series indicate features that are common between the outlet guide vane assembly **228** and the outlet guide vane assemblies **28**, **128**. The descriptions of the outlet guide vane assemblies **28**, **128** are incorporated by reference to apply to the outlet guide vane assembly **228**, except in instances when they conflict with the specific description and the drawings of the outlet guide vane assembly **228**.

(82) Similar to the outlet guide vane assemblies **28**, **128** described above, the outlet guide vane assembly **228** includes first tip and hub segments **232A**, **242A** and a central segment **248**. Unlike the assemblies described above, the outlet guide vane assembly **228** further includes a second tip segment **232B** and a second hub segment **242B** arranged radially between the first tip segment **232A** and the central segment **248** and between the first hub segment **242A** and the central segment **248**, respectively. The additional second tip segment **232B** and second hub segment **242B** allow for additional variations of the fan exit air **15** flowing over the vane **230**. In some embodiments, the central segment **248** may not be included such that the vane **230** only includes the four segments **232A**, **232B**, **242A**, **242B**.

(83) In order to move all four segments **232A**, **232B**, **242A**, **242B**, the assembly **228** includes a first actuation assembly **270** configured substantially similarly to the actuation assemblies **70**, **170**. Specifically, instead of two cams arranged on the cam rod **278**, such as the cams **54**, **57** described above, the first actuation assembly **270** includes four cams **254A**, **254B**, **257A**, **257B** each associated with a respective segment **232A**, **232B**, **242A**, **242B**, as shown in FIG. **14**. Each segment

232A, 232B, 242A, 242B includes a respective cam actuation rod 255A, 255B, 258A, 258B rotatably extending therefrom and configured to engage with a respective cam 254A, 254B, 257A, 257B. Similar to the actuation assemblies 70, 170, the cams 254A, 254B, 257A, 257B may be selectively clocked such that rotation of the cam rod 278 can cause the segments 232A, 232B, 242A, 242B to move to the same or differing pitch angles. In some embodiments, the segments 232A, 232B, 242A, 242B are rotatably hinged to each other via the same hinge and hinge receiving cavity arrangements as those described above with regard to the outlet guide vane assembly 128.

(84) Similar to the outlet guide vane assembly 128, in some embodiments, the outlet guide vane assembly 228 may include a single or multiple air manipulating members 290 arranged in the small radial gaps between the segment 232A, 232B, 242A, 242B and central segment 248. The air manipulating members 290 may be formed similarly to the air manipulating member 90, 90S, 190, 190S described above, in particular formed as seals or winglets.

(85) In some embodiments, one of the air manipulating members 290 may be formed as a winglet while the others are formed as seals. In some embodiments, one of the air manipulating members 290 may be formed as a seal while the others are formed as winglets. In some embodiments, two of the air manipulating members 290 may be formed as winglets while the others are formed as seals. In some embodiments, all of the air manipulating members 290 are formed as seals. In some embodiments, all of the air manipulating members 290 are formed as winglets.

(86) As can be seen in FIG. 15, the forward edges 296 of the winglets 290 are long enough such that both the suction and pressure side edges are located outside of the potential range of pitch angle positions of the segments 232A, 232B, 242A, 242B. This ensures that radial flows across the segments 232A, 232B, 242A, 242B will continue to be better matched at all pitch angle positions of the segments 232A, 232B, 242A, 242B.

(87) Another embodiment of an outlet guide vane assembly 328 is shown in FIG. 16. The outlet guide vane assembly 328 is similar to the outlet guide vane assemblies 28, 128, 228 shown in FIGS. 1-15 and described herein. Accordingly, similar reference numbers in the 300 series indicate features that are common between the outlet guide vane assembly 328 and the outlet guide vane assemblies 28, 128, 228. The descriptions of the outlet guide vane assemblies 28, 128, 228 are incorporated by reference to apply to the outlet guide vane assembly 328, except in instances when they conflict with the specific description and the drawings of the outlet guide vane assembly 328.

(88) The outlet guide vane assembly 328 is similar to the outlet guide vane assemblies 28, 128 described above, in particular including first tip and hub segments 332, 342 that are rotated via actuation by cam mechanisms. Unlike the outlet guide vane assemblies 28, 128, the assembly 328 does not include a single cam rod, but instead includes two actuation assemblies 370, 380 each having its own unique cam rod 378, 388. The first actuation assembly 370 is configured to rotate the first tip segment 332, and the second actuation assembly 380 is configured to rotate the first hub segment 342.

(89) As can be seen in FIG. 16, the first actuation assembly 370 includes a first actuation arm 374, a first actuation head 376 coupled to the first actuation arm 374, the first cam rod 378, and a first cam 354 fixedly coupled to the first cam rod 378. Similar to the first actuation arm 74, first actuation head 76, first cam rod 78, and first cam 54 described above, the first cam rod 378 is configured to be selectively rotated by pivoting movement of the actuation arm 374, and thus rotation of the actuation head 376, about the central axis of the cam rod 378 so as to selectively rotate the first cam 354. Rotation of the first cam 354 causes rotation of the first tip segment 332 about the leading edge pitch axis 339 to a first pitch angle relative to the incoming fan exit air 15.

(90) The components of the first actuation assembly 370, including the first actuation arm 374, the first actuation head 376, the first cam rod 378, and the first cam 354, may be arranged similarly to the first actuation arm 74, first actuation head 76, first cam rod 78, and first cam 54 described above. In particular, the first actuation assembly 370 is arranged radially outward of the fixed aft portion 350 such that the first actuation head 376 is arranged at least partially within the fan duct

outer casing **19**. The first actuation assembly **370** further includes a first cam actuation rod **355** rotatably coupled to the first tip segment **332** and extending axially aft therefrom toward the first cam **354**.

(91) An axially aft end of the first cam actuation rod **355** engages an axially forward facing surface **354C** of the first cam **354** such that rotation the first cam **354** in a first rotational direction causes the axially forward facing surface **354C** to move the first cam actuation rod **355** axially forward. The first cam actuation rod **355** is rotatably coupled to the first tip segment **332** at a point offset from the leading edge pitch axis **339** in a circumferential direction, such as within the first actuation rod receiving recess **337**. As a result, axially forward movement of the first cam actuation rod **355** rotates the first tip segment **332** in the first rotational direction.

(92) As can be seen in FIG. **16**, the second actuation assembly **380** is formed similarly to the first actuation assembly **380**. In particular the second actuation assembly **380** includes a second actuation arm **384**, a second actuation head **386** coupled to the second actuation arm **384**, the second cam rod **388**, and a second cam **357** fixedly coupled to the second cam rod **388**. The second cam rod can be radially spaced apart from the first cam rod **378** by a distance, such as the distance shown in FIG. **16**, or can be located proximate to the first cam rod **378**, nearly touching the cam rod **378**.

(93) Similar to the first cam rod **378**, the second cam rod **388** is configured to be selectively rotated by pivoting movement of the actuation arm **384**, and thus rotation of the actuation head **386**, about the central axis of the cam rod **388** so as to selectively rotate the second cam **357**. Rotation of the second cam **357** causes rotation of the first hub segment **342** about the leading edge pitch axis **339** to a second pitch angle relative to the incoming fan exit air **15**. The second pitch angle can be the same or different than the first pitch angle.

(94) The second actuation assembly **380** is arranged radially inward of the fixed aft portion **350** such that the second actuation head **386** is arranged at least partially within the inner wall **23**. The second actuation assembly **380** further includes a second cam actuation rod **358** rotatably coupled to the first hub segment **342** and extending axially aft therefrom toward the second cam **357**.

(95) An axially aft end of the second cam actuation rod **358** engages an axially forward facing surface **357C** of the second cam **357** such that rotation the second cam **357** in a first rotational direction causes the axially forward facing surface **357C** to move the second cam actuation rod **358** axially forward. The second cam actuation rod **358** is rotatably coupled to the first hub segment **342** at a point offset from the leading edge pitch axis **339** in a circumferential direction, such as within the second actuation rod receiving recess **347**. As a result, axially forward movement of the second cam actuation rod **358** rotates the first hub segment **342** in the first rotational direction.

(96) The first and second cams **354**, **357** are configured to be selectively clocked to unique rotational positions on the first and second cam rods **378**, **388** such that rotation of the first cam **354** causes the first tip segment **332** to be rotated to a first pitch angle relative to the incoming fan exit air **15** and rotation of the second cam **357** causes the first hub segment **342** to be rotated to a second angle relative to the incoming fan exit air **15**. Similar to the first actuation assembly **70**, an annular ring **362**, or annular ring segments or individually actuators, can be utilized to move the actuation arm **374**, or actuation arms **374** when there are a plurality of outlet guide vanes **330**, thus rotating the first tip segment **332**.

(97) Similar to the first actuation assembly **370**, the second actuation assembly **380** can include a second annular ring **364**, or annular ring segments or individually actuators arranged radially inward of the fan duct **24**, as shown in FIG. **16**. The second annular ring **364**, ring segments, or individual actuators may be formed and configured similarly to the first annular ring **362**, ring segments, or individual actuators described above. Circumferential movement of the second annular ring **364** moves the actuation arm **384**, or actuation arms **384** when there are a plurality of outlet guide vanes **330**, thus rotating the first hub segment **342**. In some embodiments, the cam rods **378**, **388** may be offset from the fixed aft portion **350** and extend along a side of the fixed aft

portion **350**, but in other embodiments, as shown in FIG. **16**, cam rod receiving cavities **351**, **352** may be formed in the fixed aft portion **350** through which the cam rods **378**, **388** may extend.

(98) Another embodiment of an outlet guide vane assembly **428** is shown in FIG. **17**. The outlet guide vane assembly **428** is similar to the outlet guide vane assemblies **28**, **128**, **228**, **328** shown in FIGS. **1-16** and described herein. Accordingly, similar reference numbers in the **400** series indicate features that are common between the outlet guide vane assembly **428** and the outlet guide vane assemblies **28**, **128**, **228**, **328**. The descriptions of the outlet guide vane assemblies **28**, **128**, **228**, **328** are incorporated by reference to apply to the outlet guide vane assembly **428**, except in instances when they conflict with the specific description and the drawings of the outlet guide vane assembly **428**.

(99) Similar to the outlet guide vane assembly **328** described above, the outlet guide vane assembly **428** includes first tip and hub segments **432**, **442** and two actuation assemblies **470**, **480** each associated with one of the first tip and hub segments **432**, **442**. The two actuation assemblies **470**, **480** are formed similarly to the two actuation assemblies **370**, **380** described above. Unlike the vane assembly **328** described above, the first variable leading edge outlet guide vane **430** further includes a central segment **448** arranged between the first tip segment **432** and the first hub segment **442** such that the first tip segment **432** and the first hub segment **442** are radially spaced apart. In some embodiments, the central segment **448** may be coupled to and extend axially away from the axially forward side **453** of the fixed aft portion **450**. The central segment **448** is static and does not rotate.

(100) As can be seen in FIG. **17**, the central segment **448** includes a radially outer side **448A**, a radially inner side **448B**, and an axially forward end **448C** extending between the radially outer and inner sides **448A**, **448B** and partially defining the leading edge of the vane **430** along with the tip and hub segments **432**, **442**. The central segment **448** may further include a radially outer hinge rod receiving cavity **448D** configured to receive a hinge rod **437A** of the tip segment **432**, and a radially inner hinge rod receiving cavity **448E** configured to receive a hinge rod **447A** of hub segment **442**. The central segment **448** may provide stability to the tip and hub segments **432**, **442**, as well as allow for uninterrupted flow over the central portion of the vane **430**. The first tip and hub segments **432**, **442** further include actuation rod receiving recesses **437B**, **447B** formed therein in that receive the actuation rods **455**, **458**, similar to the actuation rods **55**, **58** described above.

(101) Another embodiment of an outlet guide vane assembly **528** is shown in FIG. **18**. The outlet guide vane assembly **528** is similar to the outlet guide vane assemblies **28**, **128**, **228**, **328**, **428** shown in FIGS. **1-17** and described herein. Accordingly, similar reference numbers in the **500** series indicate features that are common between the outlet guide vane assembly **528** and the outlet guide vane assemblies **28**, **128**, **228**, **328**, **428**. The descriptions of the outlet guide vane assemblies **28**, **128**, **228**, **328**, **428** are incorporated by reference to apply to the outlet guide vane assembly **528**, except in instances when they conflict with the specific description and the drawings of the outlet guide vane assembly **528**.

(102) Similar to the outlet guide vane assemblies **328**, **428** described above, the outlet guide vane assembly **528** two actuation assemblies **570**, **580**. The two actuation assemblies **570**, **580** are formed similarly to the two actuation assemblies **370**, **380**, **470**, **480** described above. Unlike the outlet guide vane assemblies **328**, **428**, the outlet guide vane assembly **528** includes four segments **532A**, **532B**, **542A**, **542B** and a central segment **548** that is static, similar to the embodiment shown in FIG. **14** and described above. In some embodiments, the central segment **548** may be omitted such that the vane **530** only includes the four segments **532A**, **532B**, **542A**, **542B**.

(103) In order to move all four segments **532A**, **532B**, **542A**, **542B**, the first actuation assembly **570** is configured substantially similarly to the first actuation assemblies **370**, **470** described above, but instead includes two cams **554A**, **554B** arranged on the first cam rod **578** and each associated with a respective tip segment **532A**, **532B**. Similarly, the second actuation assembly **580** includes two cams **557A**, **557B** arranged on the second cam rod **588** and each associated with a respective

hub segment **542A**, **542B**.

(104) Each segment **532A**, **532B**, **542A**, **542B** includes a respective cam actuation rod **555A**, **555B**, **558A**, **558B** rotatably extending therefrom and configured to engage with a respective cam **554A**, **554B**, **557A**, **557B**. The cams **554A**, **554B**, **557A**, **557B** may be selectively clocked such that rotation of the cam rods **578**, **588** can cause the segments **532A**, **532B**, **542A**, **542B** to move to the same or differing pitch angles. In some embodiments, the segments **532A**, **532B**, **542A**, **542B** are rotatably hinged to each other via the same hinge and hinge receiving cavity arrangements as those described above with regard to the outlet guide vane assembly **128**.

(105) Another embodiment of an outlet guide vane assembly **628** is shown in FIG. **19**. The outlet guide vane assembly **628** is similar to the outlet guide vane assemblies **28**, **128**, **228**, **328**, **428**, **528** shown in FIGS. **1-18** and described herein. Accordingly, similar reference numbers in the **600** series indicate features that are common between the outlet guide vane assembly **628** and the outlet guide vane assemblies **28**, **128**, **228**, **328**, **428**, **528**. The descriptions of the outlet guide vane assemblies **28**, **128**, **228**, **328**, **428**, **528** are incorporated by reference to apply to the outlet guide vane assembly **628**, except in instances when they conflict with the specific description and the drawings of the outlet guide vane assembly **628**.

(106) The outlet guide vane assembly **628** is formed similarly to the outlet guide vane assembly **28** described above, in particular including tip and hub segments **632**, **642**. The outlet guide vane assembly **628** can include one or two actuation assemblies for controlling the segments **632**, **642**, any of which as are described above. Although FIG. **19** only shows a first actuation assembly **670** having a first actuation arm **674**, a first cam rod **678** having cams **654**, **657** attached thereto, two cam rods and two actuation arms could also be utilized as described above.

(107) As can be seen in FIG. **19**, the first actuation assembly **670** does not include cam actuation rods. Instead, the cams **654**, **657** are arranged so as to directly interact with a tab **654T**, **657T** formed on and extending axially aft of the axially aft end **634**, **644** of the segments **632**, **642**. In some embodiments, the cams **654**, **657** may be arranged within cavities **652** formed in the axial aft portion **650**, as shown in FIG. **19**. When the cams **654**, **657** are rotated via rotation of the cam rod **678** in the first rotational direction, the forward cam surface **654C**, **657C** of the respective cams **654**, **657** move against the tab **654T**, **657T** of the axially aft end **634**, **644** of the respective segment **632**, **642**, thus rotating the respective segment **632**, **642** in the second rotational direction. FIG. **19** shows various rotational positions of the cam rod **678** and cams **654**, **657**, and thus various pitch angles of the segments **632**, **642**.

(108) A method can include arranging a fan duct **20** circumferentially around a central axis **11**, providing a fan **21** comprising a plurality of fan blades **22** that extend radially outward relative to the central axis **11** and that are adapted to rotate about the central axis **11** to force fan exit air **15** toward an aft end of the fan duct **20**, and arranging an outlet guide vane assembly **28** in the fan duct **20** axially downstream of the fan **21** and configured to adjust a direction of incoming fan exit air **15** received from the plurality of fan blades **22**, the outlet guide vane assembly **28** including a first variable leading edge guide vane **30** that extends radially relative to the central axis **11** and includes a leading edge portion **31** and a fixed aft portion **50**, the leading edge portion including a first tip segment **32** configured to rotate about a leading edge pitch axis **39** and a first hub segment **42** located radially inward of and separate from the first tip segment **32**, the first hub segment **42** configured to independently rotate about the leading edge pitch axis **39** relative to the first tip segment **32**.

(109) The method can further include arranging a first actuation assembly **70** relative to the first variable leading edge guide vane **30**, the first actuation assembly **70** including a first cam rod **78**, selectively coupling a first cam **54** to the first cam rod **78**, and selectively coupling a second cam **57** to the first cam rod **78** radially spaced apart from the first cam **54**. The first cam rod **78** is configured to be selectively rotated so as to rotate the first cam **54** and the second cam **57**, wherein rotation of the first cam **54** causes rotation of the first tip segment **32** about the leading edge pitch

axis 39. Rotation of the second cam 57 causes rotation of the first hub segment 42 about the leading edge pitch axis 39. The first and second cams 54, 57 are configured to be selectively clocked to unique rotational positions on the first cam rod 78 such that rotation of the first and second cams 54, 57 causes the first tip segment 32 to be rotated to a first pitch angle relative to the incoming fan exit air 15 and the first hub segment 42 to be rotated to a second angle relative to the incoming fan exit air 15.

(110) While the disclosure has been illustrated and described in detail in the foregoing drawings and description, the same is to be considered as exemplary and not restrictive in character, it being understood that only illustrative embodiments thereof have been shown and described and that all changes and modifications that come within the spirit of the disclosure are desired to be protected. Moreover, each of the embodiments described herein, including the specific features and characteristics of each embodiment, may be combined with other embodiments as would be understood by one of ordinary skill in the art.

Claims

1. A fan assembly for a gas turbine engine, the fan assembly comprising a fan duct arranged circumferentially around a central axis, a fan comprising a plurality of fan blades that extend radially outward relative to the central axis and that are adapted to rotate about the central axis to force fan exit air toward an aft end of the fan duct, and an outlet guide vane assembly located in the fan duct axially downstream of the fan and configured to adjust a direction of incoming fan exit air received from the plurality of fan blades, the outlet guide vane assembly including a first variable leading edge outlet guide vane that extends radially relative to the central axis and includes a leading edge portion and a fixed aft portion, the leading edge portion including a first tip segment configured to rotate about a leading edge pitch axis and a first hub segment located radially inward of and separate from the first tip segment, the first hub segment configured to independently rotate about the leading edge pitch axis relative to the first tip segment, and a first actuation assembly including a first cam rod, a first cam selectively coupled to the first cam rod, and a second cam selectively coupled to the first cam rod and radially spaced apart from the first cam, wherein the first cam rod is configured to be selectively rotated so as to rotate the first cam and the second cam, wherein rotation of the first cam causes rotation of the first tip segment about the leading edge pitch axis, wherein rotation of the second cam causes rotation of the first hub segment about the leading edge pitch axis, and wherein the first and second cams are configured to be selectively clocked to unique rotational positions on the first cam rod such that rotation of the first and second cams causes the first tip segment to be rotated to a first pitch angle relative to the incoming fan exit air and the first hub segment to be rotated to a second angle relative to the incoming fan exit air.
2. The fan assembly of claim 1, wherein the first actuation assembly is configured to rotate the first tip segment and the first hub segment to the first pitch angle and the second pitch angle which is different than the first pitch angle.
3. The fan assembly of claim 2, wherein the first actuation assembly further includes a first cam actuation rod rotatably coupled to the first tip segment and extending axially aft therefrom toward the first cam, and a second cam actuation rod rotatably coupled to the first hub segment extending axially aft therefrom toward the second cam.
4. The fan assembly of claim 3, wherein an axially aft end of the first cam actuation rod engages an axially forward facing surface of the first cam such that rotation of the first cam in a first rotational direction causes the axially forward facing surface to move the first cam actuation rod axially forward, and wherein an axially aft end of the second cam actuation rod engages an axially forward facing surface of the second cam such that rotation of the second cam in the first rotational direction causes the axially forward facing surface to move the second cam actuation rod axially forward.

5. The fan assembly of claim 4, wherein the first cam actuation rod is rotatably coupled to the first tip segment at a point offset from the leading edge pitch axis in a circumferential direction such that axially forward movement of the first cam actuation rod rotates the first tip segment in the first rotational direction, and wherein the second cam actuation rod is rotatably coupled to the first hub segment at a point offset from the leading edge pitch axis in the circumferential direction such that axially forward movement of the second cam actuation rod rotates the first hub segment in the first rotational direction.
6. The fan assembly of claim 5, wherein the first tip segment includes a first cam actuation rod receiving recess formed at least partially in an aft side of the first tip segment, wherein the first cam actuation rod is rotatably mounted within the first cam actuation rod receiving recess, wherein the first hub segment includes a second cam actuation rod receiving recess formed at least partially in an aft side of the first hub segment, and wherein the second cam actuation rod is rotatably mounted within the second cam actuation rod receiving recess.
7. The fan assembly of claim 3, wherein the first actuation assembly is arranged radially outward of the fixed aft portion and includes a first actuation head, and wherein the first cam rod is fixedly coupled to the first actuation head such that rotation of the first actuation head causes rotation of the first cam rod.
8. The fan assembly of claim 7, wherein the first actuation assembly further includes a first actuation arm coupled to the first actuation head and extending generally axially aft therefrom, wherein pivoting of the first actuation arm about a rotation axis of the first actuation head causes rotation of the first actuation head which causes rotation of the first cam rod.
9. The fan assembly of claim 8, wherein the fixed aft portion includes a cam rod receiving cavity formed therethrough, and wherein the first actuation head is circumferentially aligned with the fixed aft portion such that the first cam rod extends through the cam rod receiving cavity.
10. The fan assembly of claim 8, wherein the first actuation head is circumferentially offset from the fixed aft portion such that the first cam rod radially extends circumferentially adjacent to the fixed aft portion.
11. The fan assembly of claim 8, wherein the outlet guide vane assembly further includes a first annular ring extending circumferentially about the central axis, wherein an axially aft end of the first actuation arm is connected to the first annular ring, and wherein circumferential movement of the first annular ring causes movement of the first actuation arm generally circumferentially relative to the first actuation head which causes the first actuation arm to pivot about the rotation axis of the first actuation head which causes rotation of the first actuation head which causes rotation of the first cam rod.
12. A fan assembly for a gas turbine engine, the fan assembly comprising a fan duct arranged circumferentially around a central axis, a fan adapted to rotate about the central axis to force fan exit air toward an aft end of the fan duct, and an outlet guide vane assembly located in the fan duct axially downstream of the fan and configured to adjust a direction of incoming fan exit air received from the plurality of fan blades, the outlet guide vane assembly including a first variable leading edge outlet guide vane including a leading edge portion and a fixed aft portion, the leading edge portion including a first tip segment configured to rotate about a leading edge pitch axis, and a first actuation assembly including a first cam rod extending through the fixed aft portion and a first cam selectively coupled to the first cam rod, wherein the first cam rod is configured to be selectively rotated so as to rotate the first cam, wherein rotation of the first cam causes rotation of the first tip segment about the leading edge pitch axis, and wherein the first cam is configured to be selectively clocked to rotational positions on the first cam rod such that rotation of the first cam causes the first tip segment to be rotated to a first pitch angle relative to the incoming fan exit air.
13. The fan assembly of claim 12, wherein the leading edge portion further includes a first hub segment located radially inward of and separate from the first tip segment, the first hub segment configured to independently rotate about the leading edge pitch axis relative to the first tip segment.

14. The fan assembly of claim 13, wherein the first actuation assembly further includes a second cam selectively coupled to the first cam rod and radially spaced apart from the first cam, wherein the first cam rod is configured to be selectively rotated so as to rotate the first cam and the second cam, and wherein the first cam and the second cam are configured to be selectively clocked to unique rotational positions on the first cam rod such that rotation of the first and second cams causes the first tip segment to be rotated to a first pitch angle relative to the incoming fan exit air and the first hub segment to be rotated to a second angle relative to the incoming fan exit air.

15. The fan assembly of claim 14, wherein the first actuation assembly further includes a first cam actuation rod rotatably coupled to the first tip segment and extending axially aft therefrom toward the first cam, and a second cam actuation rod rotatably coupled to the first hub segment extending axially aft therefrom toward the second cam.

16. The fan assembly of claim 15, wherein an axially aft end of the first cam actuation rod engages an axially forward facing surface of the first cam such that rotation of the first cam in a first rotational direction causes the axially forward facing surface to move the first cam actuation rod axially forward, and wherein an axially aft end of the second cam actuation rod engages an axially forward facing surface of the second cam such that rotation of the second cam in the first rotational direction causes the axially forward facing surface to move the second cam actuation rod axially forward.

17. The fan assembly of claim 16, wherein the first cam actuation rod is rotatably coupled to the first tip segment at a point offset from the leading edge pitch axis in a circumferential direction such that axially forward movement of the first cam actuation rod rotates the first tip segment in the first rotational direction, and wherein the second cam actuation rod is rotatably coupled to the first hub segment at a point offset from the leading edge pitch axis in the circumferential direction such that axially forward movement of the second cam actuation rod rotates the first hub segment in the first rotational direction.

18. A method comprising arranging a fan duct circumferentially around a central axis, providing a fan comprising a plurality of fan blades that extend radially outward relative to the central axis and that are adapted to rotate about the central axis to force fan exit air toward an aft end of the fan duct, arranging an outlet guide vane assembly in the fan duct axially downstream of the fan and configured to adjust a direction of incoming fan exit air received from the plurality of fan blades, the outlet guide vane assembly including a first variable leading edge guide vane that extends radially relative to the central axis and includes a leading edge portion and a fixed aft portion, the leading edge portion including a first tip segment configured to rotate about a leading edge pitch axis and a first hub segment located radially inward of and separate from the first tip segment, the first hub segment configured to independently rotate about the leading edge pitch axis relative to the first tip segment, arranging a first actuation assembly relative to the first variable leading edge guide vane, the first actuation assembly including a first cam rod, selectively coupling a first cam to the first cam rod, and selectively coupling a second cam to the first cam rod radially spaced apart from the first cam, wherein the first cam rod is configured to be selectively rotated so as to rotate the first cam and the second cam, wherein rotation of the first cam causes rotation of the first tip segment about the leading edge pitch axis, wherein rotation of the second cam causes rotation of the first hub segment about the leading edge pitch axis, and wherein the first and second cams are configured to be selectively clocked to unique rotational positions on the first cam rod such that rotation of the first and second cams causes the first tip segment to be rotated to a first pitch angle relative to the incoming fan exit air and the first hub segment to be rotated to a second angle relative to the incoming fan exit air.
